

A nearly sharp rational quadrature upper bound from Mit'kin

A consequence of published Hilbert–Kamke and real quadrature results

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Put

$$H(r) = \sum_{a=0}^{\lfloor \log_2 r \rfloor} 2^a (2^{\lfloor r/2^a \rfloor} - 1) = 2^r + O(2^{r/2}).$$

Let R_r be the minimum size of a rational equal-weight uniform-interval quadrature rule of degree r , allowing repeated nodes.

Theorem 1. *For every sufficiently large r , every integer $K \geq H(r)$ admits rational nodes $x_1, \dots, x_K \in (0, 1)$ with*

$$K^{-1} \sum_i x_i^j = 1/(j+1) \quad (0 \leq j \leq r).$$

Thus $R_r \leq 2^r + O(2^{r/2})$. Together with the independently proved optimized parity-filter bound $R_r \geq 2^{2\lfloor r/2 \rfloor}$, this gives $2^r \leq R_r \leq 2^r + O(2^{r/2})$ for even r .

The upper bound is derived from earlier published results; no historical novelty claim is made. The only adaptation is to keep the circle-method summation interval fixed at $[0, P]$, rather than tying its endpoint to a prescribed power sum. Unlocalized positive-integer solutions alone would not ensure rational nodes in $[0, 1]$.

Lemma 2 (Mit'kin's uniform local input). *Let $r \geq 12$, $K \geq H(r)$, and $V = (h^j)_{1 \leq j, h \leq r}$. If $N \in \mathbb{Z}^r$ and $Vt = N$ has an integer solution, then the Hilbert–Kamke singular series $\mathfrak{S}_{K,r}(N)$ is bounded below by a positive constant depending only on r, K .*

Source and precise extraction. Mit'kin [1, Lemma 15, printed p. 577] proves uniform positivity whenever $K \geq R(r) \geq r^2 + 2$ and all local lifting thresholds $\tilde{r}(r, p)$ are at most $R(r)$. The estimates assembled on printed pp. 589–590 below the main theorem give $\tilde{r}(r, p) \leq H(r)$ for every prime p when $r \geq 12$. The $p = 2$ bound is the corollary to Proposition 2 on p. 589; the remaining bounds in (94) are at most $H(r)$. The arithmetic condition is precisely integer solvability of $Vt = N$, as stated in the introduction and reformulated in (13)–(15). Also $H(r) \geq r^2 + 2$ in this range. $\square$

Lemma 3 (Fixed-interval circle method). *Fix $r \geq 3$ and $K \geq r(r+1) + 1$. Let $c \in \mathbb{Q}^r$ be fixed, and let P tend to infinity through integers for which $N_j = c_j P^j$ are integers. The number $A(P, N)$ of solutions*

$$1 \leq y_i \leq P-1, \quad \sum_i y_i^j = N_j \quad (1 \leq j \leq r)$$

satisfies, with $D = r(r+1)/2$,

$$A(P, N) = P^{K-D} (\mathfrak{S}_{K,r}(N) \mathfrak{J}_{K,r}(c) + o(1)),$$

where

$$\mathfrak{J}_{K,r}(c) = \int_{\mathbb{R}^r} \left(\int_0^1 e(\beta_1 t + \cdots + \beta_r t^r) dt \right)^K e(-\beta \cdot c) d\beta.$$

If the real moment system has an interior solution with r distinct coordinates, then $\mathfrak{J}_{K,r}(c) > 0$.

Proof. We extract the standard calculation from [2, Sections 3–6], keeping the endpoint P independent of the targets. Put

$$f(\alpha) = \sum_{1 \leq y \leq P-1} e(\alpha_1 y + \cdots + \alpha_r y^r).$$

By orthogonality $A(P, N) = \int_{[0,1]^r} f(\alpha)^K e(-\alpha \cdot N) d\alpha$. The target factor has modulus one in all the following estimates.

Take the major arcs $\mathfrak{P} = \mathfrak{K}(L)$ of [2, Section 3], with $L = P^{1/(8r^2)}$: they have $q \leq L$, $(q, a_1, \dots, a_r) = 1$, and $|\alpha_j - a_j/q| \leq LP^{-j}$ for all j . Lemma 4.1 of that paper gives

$$\sup_{\alpha \notin \mathfrak{P}} |f(\alpha)| \ll P^{1-\delta}, \quad \delta = 1/(12r^3).$$

Deleting the two endpoints changes the sum by at most two. The Vinogradov mean value theorem, as in (1.3) of [2], gives

$$\int_{[0,1]^r} |f(\alpha)|^{2D} d\alpha \ll_\epsilon P^{D+\epsilon}.$$

Restricting the summation interval only decreases this even moment, which counts solutions. Hence the absolute minor-arc contribution is

$$\ll P^{K-D-\delta(K-2D)+\epsilon} = o(P^{K-D}),$$

with $0 < \epsilon < \delta(K - 2D)$.

On a major arc, the approximation of [2, Section 6] is

$$f(\alpha) = q^{-1} S(q, a) I(\alpha - a/q; P) + O(L^2),$$

where $S(q, a) = \sum_{h=1}^q e((a_1 h + \cdots + a_r h^r)/q)$ and $I(\beta; P) = \int_0^P e(\beta_1 t + \cdots + \beta_r t^r) dt$. The major arcs have total measure $O(L^{2r+1} P^{-D})$, so replacement of f^K has total error

$$O(P^{K-D-1} L^{2r+3}) = o(P^{K-D}).$$

The remaining expression is the truncated singular series times the truncated singular integral. The estimates used in [2, Section 6],

$$|S(q, a)| \ll_\epsilon q^{1-1/r+\epsilon}, \quad |I(\beta; 1)| \ll (1 + |\beta_1| + \cdots + |\beta_r|)^{-1/r},$$

give uniformly negligible tails: absolute convergence follows from $K > r(r+1)$ for the series and $K > r^2$ for the integral. The change $\beta_j \mapsto P^{-j} \beta_j$ leaves $c_j = N_j/P^j$ fixed, proving the asymptotic. None of these steps requires P to equal the largest root of a target moment.

At an interior real solution with r distinct coordinates, the moment Jacobian has an invertible Vandermonde minor. The implicit function theorem supplies a positive density at c from a small box about that solution. Fourier inversion, justified by absolute convergence, identifies the whole nonnegative density with $\mathfrak{J}_{K,r}(c)$, proving strict positivity. $\square$

Proof of Theorem 1. Set $c_j = K/(j+1)$, $B = |\det V|$, and restrict P to multiples of $B(r+1)!$. Then every $N_j = KP^j/(j+1)$ is an integer divisible by B , so $V^{-1}N = (\operatorname{adj} V)N/\det V$ is integral. Lemma 2 gives a uniform positive singular-series lower bound.

For sufficiently large r , $H(r)$ exceeds both $r(r+1)+1$ and the quadratic node threshold in [3] for a real equal-weight rule of degree $2r+2$. Such a rule has at least $r+2$ distinct support points: otherwise the square of a polynomial vanishing on its support contradicts exactness. At least r support points are interior. Fix these as pivots, move all endpoint coordinates slightly inward, and use the implicit function theorem on the first r moments to correct the pivots. This gives an interior real solution with nonsingular Jacobian. Thus $\mathfrak{J}_{K,r}(c) > 0$.

Lemma 3 now gives $A(P, N) > 0$ for sufficiently large multiples P . The rational numbers $x_i = y_i/P$ lie in $(0, 1)$ and have the required moments. $\square$

Checks and scope. Mit'kin's Lemma 15 and final theorem were checked visually in the original English PDF, printed pp. 577 and 589. The circle-method lemma is an explicit deduction from the cited estimates, not a quotation of Wooley's Theorem 1.1. No common-denominator bound is asserted. This concerns scalar rational quadrature and does not close the exponential versus factorial-size gap for permutation-fair dice.

References

- [1] D. A. Mit'kin, *An estimate for the number of terms in the Hilbert–Kamke problem*, Math. USSR-Sbornik 57 (1987), 561–590, doi:10.1070/SM1987v057n02ABEH003087.
- [2] Trevor D. Wooley, *Subconvexity and the Hilbert–Kamke problem*, Math. Z. (2025), doi:10.1007/s00209-025-03812-9; author's full text.
- [3] Shoni Gilboa and Ron Peled, *Chebyshev-type quadratures for doubling weights*, Constructive Approximation 45 (2017), 193–216, arXiv:1507.01505.